

MOTS-c

Mitochondrial Open Reading Frame of the 12S rRNA-c

MITOCHONDRIAL	FAT LOSS	INSULIN SENSITIVITY	LONGEVITY	EXERCISE MIMETIC
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WHAT IS IT?

MOTS-c is a mitochondrial-derived peptide (MDP) encoded within the 12S ribosomal RNA gene of mitochondrial DNA. First identified in 2015 by Lee et al. at USC, it functions as a systemic metabolic regulator — improving insulin sensitivity, promoting fat utilization, enhancing mitochondrial function, and extending lifespan in animal models. It declines significantly with age, making it a compelling longevity and metabolic compound.

DOSING AT A GLANCE

Weekly SC dose	Daily SC (alt. protocol)	Optimal timing	Cycle
5–15 mg	1–5 mg	Morning	8–12 weeks

KEY BENEFITS

- Activates AMPK — the master metabolic switch governing energy utilization
- Improves insulin sensitivity — reduces insulin resistance in skeletal muscle
- Promotes fat oxidation and lipolysis — shifts fuel use toward stored fat
- Enhances mitochondrial biogenesis and function
- Exercise mimetic — replicates some benefits of aerobic exercise without movement
- Extends healthy lifespan in mouse models
- Anti-inflammatory — reduces oxidative stress systemically
- Improves skeletal muscle function and endurance capacity
- Cognitive protection — crosses blood-brain barrier; neuroprotective effects
- Reduces age-related muscle loss (sarcopenia) in aging models

HOW IT WORKS

MOTS-c acts in a AMPK-dependent manner to activate the folate cycle and one-carbon metabolism, leading to AICAR accumulation — a natural AMPK activator. This mirrors the metabolic effects of exercise and caloric restriction. It also translocates to the nucleus under stress, regulating nuclear gene expression and the adaptive response to metabolic challenge.

PROTOCOLS

- **Weekly Protocol:** 10 mg SC once weekly, morning fasted.
- **Daily Protocol:** 2–5 mg SC daily, cycle 8 weeks on / 4 weeks off.
- **Exercise Stack:** Dose pre-training — synergistic with aerobic exercise for metabolic adaptation.
- **Longevity Stack:** MOTS-c + Semax + NMN/NR for comprehensive mitochondrial and cognitive support.

SIDE EFFECTS

- Generally well tolerated — no significant adverse effects in current human use
- Injection site reactions (mild, transient)
- Mild fatigue or lightheadedness at higher doses (AMPK activation effect)
- Hypoglycemia possible if combined with insulin or diabetes medications

STORAGE

Lyophilized: refrigerate or -20°C freezer. Reconstituted: refrigerate, use within 30 days.

■ **Research peptide — no human clinical trials completed. Source from reputable suppliers with HPLC certification.**
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